

SOME CHARACTERIZATIONS OF THE MATRIX VARIATE NORMAL DISTRIBUTION

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Summary: In this paper some characterizations are given of the matrix variate normal distribution where the marginal matrix variates are identically distributed. These characterizations are based on

- (a) the conditional normality of one of the matrix variates given the other or on
- (b) distributional properties of a linear function of matrix or vector variates whether normality is assumed or not.

1. INTRODUCTION

Consider the matrix variate

$$\mathbf{Z} : 2p \times n = \begin{bmatrix} \mathbf{X} : p \times n \\ \mathbf{Y} : p \times n \end{bmatrix}$$

where

$$\mathbf{X} : p \times n = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{p1} & x_{p2} & \cdots & x_{pn} \end{bmatrix}$$

and

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$$\mathbf{Y} : p \times n = \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & y_{2n} \\ \vdots & \vdots & & \vdots \\ y_{p1} & y_{p2} & \cdots & y_{pn} \end{bmatrix}$$

are not necessarily independent. Le Roux and Roux (1984) show that if the marginal distributions of $\mathbf{x} : p \times 1$ and $\mathbf{y} : p \times 1$ are normal, it does not imply that the distribution of $\mathbf{z} : 2p \times 1$ is normal. For a marginal normal distribution of $\mathbf{x} : p \times 1$ and a conditional normal distribution of $\mathbf{x} : p \times 1$ given $\mathbf{y} : p \times 1$ with specific parameters, $\mathbf{z} : 2p \times 1$ can be characterized as a multivariate normal distribution (cf. Roux (1971)). Bekker, Rautenbach and Roux (1988) discuss a result and apply it in characterizing a matrix variate normal distribution in terms of two conditional normal distributions. In this paper some characterization results of the matrix variate normal distribution are derived which are based on the conditional normality of $\mathbf{Y} : p \times n$ given $\mathbf{X} : p \times n$ or on the normality of a linear function of $\mathbf{X}$ and $\mathbf{Y}$. Lastly a characterization of the multivariate normal distribution is examined using a linear function of $\mathbf{x} : p \times 1$ and $\mathbf{y} : p \times 1$ without making any assumptions regarding normality.

Suppose $\mathbf{Z} \sim MTN(2p, n, \mathbf{M} = (\mu_{ij}), \mathbf{\Sigma} \otimes \mathbf{I})$ then the probability density function (p.d.f.) of $\mathbf{Z} = (z_{ij})$ is given by (Patil, *et al.* (1984))

$$f(\mathbf{Z}) = (2\pi)^{-pn} |\mathbf{\Sigma}|^{-\frac{1}{2}n} \text{etr} \left\{ -\frac{1}{2} \mathbf{\Sigma}^{-1} (\mathbf{Z} - \mathbf{M})(\mathbf{Z} - \mathbf{M})' \right\}$$

for $-\infty < z_{ij} < \infty, -\infty < \mu_{ij} < \infty, \mathbf{\Sigma} > \mathbf{O}, n \geq 2p$.

The characteristic function of $\mathbf{Z}$ is

$$\phi(\mathbf{U}) = \text{etr}(i\mathbf{M}\mathbf{U}' - \frac{1}{2}\mathbf{\Sigma}\mathbf{U}\mathbf{U}').$$

For

$$\mathbf{U} : 2p \times n = \begin{bmatrix} \mathbf{S} : p \times n \\ \mathbf{T} : p \times n \end{bmatrix}$$

the matrices of the carriers of $\mathbf{Z}, \mathbf{X}$ and $\mathbf{Y}$ respectively, the joint characteristic function of $\mathbf{X}$ and $\mathbf{Y}$ can be written as

$$\psi(\mathbf{S}, \mathbf{T}) = \text{etr} \left\{ i\mathbf{M}_1 \mathbf{S}' + i\mathbf{M}_2 \mathbf{T}' - \frac{1}{2} (\mathbf{\Sigma}_{11} \mathbf{S} \mathbf{S}' + 2\mathbf{\Sigma}_{12} \mathbf{T} \mathbf{S}' + \mathbf{\Sigma}_{22} \mathbf{T} \mathbf{T}') \right\}$$

where

$$\mathbf{M} : 2p \times n = \begin{bmatrix} \mathbf{M}_1 : p \times n \\ \mathbf{M}_2 : p \times n \end{bmatrix} \text{ and } \boldsymbol{\Sigma} : 2p \times 2p = \begin{bmatrix} \boldsymbol{\Sigma}_{11} : p \times p & \boldsymbol{\Sigma}_{12} : p \times p \\ \boldsymbol{\Sigma}_{21} : p \times p & \boldsymbol{\Sigma}_{22} : p \times p \end{bmatrix}$$

2. CHARACTERIZATION RESULTS

THEOREM 2.1

Let $\mathbf{X} : p \times n$ and $\mathbf{Y} : p \times n$ be identically distributed matrix variates. Suppose $\mathbf{Y}|\mathbf{X} \sim MTN(p, n, a\mathbf{X} + \mathbf{B}, \boldsymbol{\Sigma} \otimes \mathbf{I}_n)$ with $\mathbf{B} : p \times n$ and $|a| < 1$. Then $\mathbf{X}$ and $\mathbf{Y}$ each has a $MTN(p, n, (1-a)^{-1}\mathbf{B}, (1-a^2)^{-1}\boldsymbol{\Sigma} \otimes \mathbf{I}_n)$ distribution and $\mathbf{Z}$ has a $MTN\left(2p, n, \mathbf{M}, (1-a^2)^{-1} \begin{bmatrix} \boldsymbol{\Sigma} & a\boldsymbol{\Sigma} \\ a\boldsymbol{\Sigma} & \boldsymbol{\Sigma} \end{bmatrix} \otimes \mathbf{I}_n\right)$ distribution with $\mathbf{M} : 2p \times n = (1-a)^{-1} \begin{bmatrix} \mathbf{B} \\ \mathbf{B} \end{bmatrix}$.

Proof

The joint characteristic function of $\mathbf{X}$ and $\mathbf{Y}$ is given by

$$\begin{aligned} (2.1) \quad \psi(\mathbf{S}, \mathbf{T}) &= E \text{etr}(i\mathbf{S}'\mathbf{X} + i\mathbf{T}'\mathbf{Y}) \\ &= E[\text{etr}(i\mathbf{S}'\mathbf{X})E_{\mathbf{Y}|\mathbf{X}}\text{etr}(i\mathbf{T}'\mathbf{Y})] \\ &= E[\text{etr}(i\mathbf{S}'(\mathbf{X} + a\mathbf{T}))\text{etr}(i\mathbf{BT}' - \frac{1}{2}\boldsymbol{\Sigma}\mathbf{T}\mathbf{T}')] \\ &= \phi(\mathbf{S} + a\mathbf{T})\text{etr}(i\mathbf{BT}' - \frac{1}{2}\boldsymbol{\Sigma}\mathbf{T}\mathbf{T}') \text{ where } \psi(\cdot) \text{ indicates} \\ &\quad \text{the characteristic function of } \mathbf{X} \text{ or } \mathbf{Y}. \end{aligned}$$

Because $\mathbf{X}$ and $\mathbf{Y}$ are identically distributed, it follows that

$$(2.2) \quad \phi(\mathbf{T}) = \psi(\mathbf{T}, \mathbf{O}) = \psi(\mathbf{O}, \mathbf{T}) = \phi(a\mathbf{T})\text{etr}(i\mathbf{BT}' - \frac{1}{2}\boldsymbol{\Sigma}\mathbf{T}\mathbf{T}').$$

It can be shown that

$$(2.3) \quad \begin{aligned} \phi(\mathbf{T}) &= \phi(a^n\mathbf{T})\text{etr}[i\mathbf{BT}'(1+a+\dots+a^{n-1}) \\ &\quad - \frac{1}{2}\boldsymbol{\Sigma}\mathbf{T}\mathbf{T}'(1+a^2+\dots+a^{2(n-1)})]. \end{aligned}$$

For $|a| < 1$ and $n \rightarrow \infty$

$$(2.4) \quad \phi(\mathbf{T}) = \text{etr} \left[i(1-a)^{-1} \mathbf{B} \mathbf{T}' - \frac{1}{2} (1-a^2)^{-1} \Sigma \mathbf{T} \mathbf{T}' \right].$$

Therefore $\mathbf{X}$ and $\mathbf{Y}$ each has a $MTN(p, n, (1-a)^{-1} \mathbf{B}, (1-a^2)^{-1} \Sigma \otimes \mathbf{I}_n)$ distribution.

By substituting (2.4) into (2.1)

$$\begin{aligned} \psi(\mathbf{S}, \mathbf{T}) = & \text{etr} \{ i(1-a)^{-1} \mathbf{B} \mathbf{S}' + i(1-a)^{-1} \mathbf{B} \mathbf{T}' \\ & - \frac{1}{2} (1-a^2)^{-1} [\Sigma \mathbf{S} \mathbf{S}' + 2a \Sigma \mathbf{T} \mathbf{S}' + \Sigma \mathbf{T} \mathbf{T}'] \} \end{aligned}$$

and it follows that $\mathbf{Z}$ has the above distribution.

REMARK 2.1

- (i) If $|a| \geq 1$ then $\mathbf{X}$ and $\mathbf{Y}$ have a degenerate distribution which is contradictory to the assumption of a conditional distribution of $\mathbf{Y}$ given $\mathbf{X}$ (cf. Ahsanullah (1985)).
- (ii) The conditions of theorem 2.1 can also be written in a slightly different but equivalent way: Let $\mathbf{X} : p \times n$ and $\mathbf{Y} : p \times n$ be identically distributed matrix variates and suppose $\mathbf{X}$ and $\mathbf{V} = \mathbf{Y} - a\mathbf{X}$ are independent matrix variates with $\mathbf{V} \sim MTN(p, n, \mathbf{B}, \Sigma \otimes \mathbf{I}_n)$, $|a| < 1$.
- (iii) For $n = 1$ theorem 2.1 holds for two identically distributed vector variates and the corresponding bivector variate normal distribution is characterized.
- (iv) For $n = 1$ and $p = 1$ the bivariate normal distribution is similarly characterized by Ahsanullah (1985).

In the following theorem multiple linear regression is assumed in the conditional normality.

THEOREM 2.2

Suppose $\mathbf{X} : p \times n$ and $\mathbf{Y} : p \times n$ are identically distributed matrix variates and that $\mathbf{Y}|\mathbf{X} \sim MTN(p, n, \mathbf{A}\mathbf{X} + \mathbf{B}, \mathbf{I}_p \otimes \mathbf{I}_n)$ with $\mathbf{B} : p \times n$, $\mathbf{A} : p \times p$ and symmetric. Let $\lambda_1, \dots, \lambda_p$ be the characteristic roots of $\mathbf{A}$ where $|\lambda_i| < 1, i = 1, \dots, p$. Then $\mathbf{X}$ and $\mathbf{Y}$ each has a $MTN(p, n, (\mathbf{I} - \mathbf{A})^{-1} \mathbf{B}, (\mathbf{I} - \mathbf{A}^2)^{-1} \otimes \mathbf{I}_n)$ distribution and

$$\mathbf{Z} \sim MTN \left(2p, n, \mathbf{M}, \begin{bmatrix} (\mathbf{I} - \mathbf{A}^2)^{-1} & \mathbf{A}(\mathbf{I} - \mathbf{A}^2)^{-1} \\ \mathbf{A}(\mathbf{I} - \mathbf{A}^2)^{-1} & (\mathbf{I} - \mathbf{A}^2)^{-1} \end{bmatrix} \otimes \mathbf{I}_n \right) \text{ with}$$

$$\mathbf{M} : 2p \times n = \begin{bmatrix} (\mathbf{I} - \mathbf{A})^{-1} & \mathbf{B} \\ (\mathbf{I} - \mathbf{A})^{-1} & \mathbf{B} \end{bmatrix}.$$

Proof

As in (2.1) the joint characteristic function of $\mathbf{X}$ and $\mathbf{Y}$ can be written as

$$(2.5) \quad \psi(\mathbf{S}, \mathbf{T}) = \phi(\mathbf{S} + \mathbf{AT}) \text{etr}(i\mathbf{BT}' - \frac{1}{2}\mathbf{TT}') \text{ where } \phi(\cdot)$$

denotes the characteristic function of $\mathbf{X}$ or $\mathbf{Y}$.

Then

$$(2.6) \quad \phi(\mathbf{T}) = \psi(0, \mathbf{T}) = \phi(\mathbf{AT}) \text{etr}(i\mathbf{BT}' - \frac{1}{2}\mathbf{TT}')$$

because $\mathbf{X}$ and $\mathbf{Y}$ are identically distributed.

Similarly to (2.3) it now follows that

$$\begin{aligned} \phi(\mathbf{T}) &= \phi(\mathbf{A}^n \mathbf{T}) \text{etr}[i\mathbf{BT}'(\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \dots + \mathbf{A}^{n-1})] \\ &\quad \cdot \text{etr}[-\frac{1}{2}\mathbf{TT}'(\mathbf{I} + \mathbf{A}^2 + \mathbf{A}^4 + \dots + \mathbf{A}^{2(n-1)})]. \end{aligned}$$

For $|\lambda_i| < 1$ and $n \rightarrow \infty$

$$(2.7) \quad \phi(\mathbf{T}) = \text{etr}[i\mathbf{BT}'(\mathbf{I} - \mathbf{A})^{-1} - \frac{1}{2}\mathbf{TT}'(\mathbf{I} - \mathbf{A}^2)^{-1}].$$

Therefore $\mathbf{X}$ and $\mathbf{Y}$ each has a $MTN(p, n, (\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}, (\mathbf{I} - \mathbf{A}^2)^{-1} \otimes \mathbf{I}_n)$ distribution.

Hence it follows from (2.5) and (2.7) that

$$\begin{aligned} \psi(\mathbf{S}, \mathbf{T}) &= \text{etr}\{i(\mathbf{I} - \mathbf{A})^{-1}\mathbf{BS}' + i(\mathbf{I} - \mathbf{A})^{-1}\mathbf{BT}' - \frac{1}{2}[(\mathbf{I} - \mathbf{A}^2)^{-1}\mathbf{SS}' \\ &\quad + 2\mathbf{A}(\mathbf{I} - \mathbf{A}^2)^{-1}\mathbf{ST}' + (\mathbf{I} - \mathbf{A}^2)^{-1}\mathbf{TT}']\}. \end{aligned}$$

REMARK 2.2

For $n = 1$ theorem 2.2 holds for two identically distributed vector variates. A further interesting characterization of the multivariate normal distribution, also based on linearity and normality, is examined in theorem 2.3. This result can easily be extended to matrix variates.

THEOREM 2.3

Suppose $\mathbf{x} : p \times 1$ and $\mathbf{v} : p \times 1$ are independent vector variates each with a $MN(p, 0, \Sigma)$ distribution. For all real a let $(\mathbf{x}, a\mathbf{x} + \mathbf{v})$ and $(\mathbf{x}, \mathbf{y})$ be identically distributed. Then $\mathbf{z} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$ has a $MN\left(2p, 0, \begin{bmatrix} \Sigma & a\Sigma \\ a\Sigma & (1+a^2)\Sigma \end{bmatrix}\right)$ distribution.

Proof

The joint characteristic function of $\mathbf{x}$ and $\mathbf{y}$ is

$$\begin{aligned} \psi(\mathbf{s}, \mathbf{t}) &= Ee^{i\mathbf{s}'\mathbf{x} + i\mathbf{t}'\mathbf{y}} \\ &= Ee^{i\mathbf{s}'\mathbf{x} + i\mathbf{t}'(a\mathbf{x} + \mathbf{v})} \\ &= Ee^{i\mathbf{x}'(\mathbf{s} + a\mathbf{t})} Ee^{i\mathbf{v}'\mathbf{t}} \\ &= \exp\left\{-\frac{1}{2}[\mathbf{s}'\Sigma\mathbf{s} + 2a\mathbf{s}'\Sigma\mathbf{t} + (1+a^2)\mathbf{t}'\Sigma\mathbf{t}]\right\} \end{aligned}$$

from which it follows that $\mathbf{z}$ has the above distribution.

Ahsanullah (1974) gives a characterization of the bivariate normal distribution without assuming normality. Subsequently a characterization is examined of a multivariate normal distribution without any assumptions regarding normality.

THEOREM 2.4

Suppose $\mathbf{x} : p \times 1$ and $\mathbf{y} : p \times 1$ are identically distributed vector variates with $E(\mathbf{x}) = E(\mathbf{y}) = 0$ and a covariance structure of the form $\begin{bmatrix} \mathbf{I}_p & \mathbf{D}_a \\ \mathbf{D}_a & \mathbf{I}_p \end{bmatrix}$ where $\mathbf{D}_a = \text{diag}(a_1, \dots, a_p)$. Then $\mathbf{z} : 2p \times 1 = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$ has a multivariate normal distribution if, and only if, $\mathbf{x}$ and $\mathbf{v}$ are independent and identically distributed where $\mathbf{v}_i = (1 - a_i^2)^{-\frac{1}{2}}(\mathbf{y}_i - a_i \mathbf{x}_i)$ for all $a_i (|a_i| < 1)$ and $i = 1, 2, \dots, p$.

Proof

Firstly suppose $\mathbf{z} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \sim MN \left(2p, 0, \begin{bmatrix} \mathbf{I}_p & \mathbf{D}_a \\ \mathbf{D}_a & \mathbf{I}_p \end{bmatrix} \right)$.

Then the joint p.d.f. of $\mathbf{x}$ and $\mathbf{v}$ is

$$\begin{aligned} f(\mathbf{x}, \mathbf{v}) &= f(\mathbf{x}, \mathbf{y}(\mathbf{x}, \mathbf{v})) |J(\mathbf{y} \rightarrow \mathbf{v})| \text{ where } J(\mathbf{y} \rightarrow \mathbf{v}) = \prod_{i=1}^p (1 - a_i^2)^{\frac{1}{2}} \\ &= (2\pi)^{-p} \exp \left[-\frac{1}{2} (\mathbf{x}'\mathbf{x} + \mathbf{v}'\mathbf{v}) \right]. \end{aligned}$$

Hence it follows that $\mathbf{x}$ and $\mathbf{v}$ are independent and identically (normally) distributed vector variates.

As for the converse, the joint characteristic function of $\mathbf{x}$ and $\mathbf{y}$ is

$$\begin{aligned} \psi(\mathbf{s}, \mathbf{t}) &= E \exp[i\mathbf{s}'\mathbf{x} + i\mathbf{t}'\mathbf{y}] \\ &= E \exp \left[i\mathbf{x}'(\mathbf{s} + \mathbf{D}_a \mathbf{t}) + i\mathbf{v}'\mathbf{D}_{(1-a^2)^{\frac{1}{2}}} \mathbf{t} \right] \end{aligned}$$

$$\text{where } \mathbf{D}_{(1-a^2)^{\frac{1}{2}}} = \text{diag}((1 - a_1^2)^{\frac{1}{2}}, \dots, (1 - a_p^2)^{\frac{1}{2}}).$$

Thus

$$(2.8) \quad \psi(\mathbf{s}, \mathbf{t}) = \phi(\mathbf{s} + \mathbf{D}_a \mathbf{t}) \phi(\mathbf{D}_{(1-a^2)^{\frac{1}{2}}} \mathbf{t})$$

because $\mathbf{x}$ and $\mathbf{v}$ are independent and identically distributed with characteristic function $\phi(\cdot)$.

From (2.8) and since $\mathbf{x}$ and $\mathbf{y}$ are identically distributed, it now follows that

$$(2.9) \quad \phi(\mathbf{t}) = \psi(0, \mathbf{t}) = \phi(\mathbf{D}_a \mathbf{t}) \phi(\mathbf{D}_{(1-a^2)^{\frac{1}{2}}} \mathbf{t}).$$

Differentiate (2.9) with regard to $\mathbf{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_p \end{bmatrix}$ and after simplification

(Bargmann, 1971)

$$(2.10) \quad \phi(\mathbf{D}_{(1-a^2)^{\frac{1}{2}}} \mathbf{t}) \phi'(\mathbf{D}_a \mathbf{t}) = \mathbf{D}_{a(1-a^2)^{-\frac{1}{2}}} \phi(\mathbf{D}_a \mathbf{t}) \phi'(\mathbf{D}_{(1-a^2)^{\frac{1}{2}}} \mathbf{t})$$

where

$$\phi'(D_a t) = \begin{bmatrix} \frac{\partial}{\partial a_1 t_1} \\ \vdots \\ \frac{\partial}{\partial a_p t_p} \end{bmatrix} \phi(D_a t) \text{ and}$$

$$\phi'(D_{(1-a^2)^{\frac{1}{2}}} t) = \begin{bmatrix} \frac{\partial}{\partial (1-a_1^2)^{\frac{1}{2}} t_1} \\ \vdots \\ \frac{\partial}{\partial (1-a_p^2)^{\frac{1}{2}} t_p} \end{bmatrix} \phi(D_{(1-a^2)^{\frac{1}{2}}} t), \text{ for all } t_i (i = 1, 2, \dots, p),$$

because $\phi(\cdot)$ and $\phi'(\cdot)$ are continuous.

Rewrite (2.10) as follows

$$(2.11) \quad \frac{\phi'(D_a t)}{\phi(D_a t)} = D_{a(1-a^2)^{-\frac{1}{2}}} \frac{\phi'(D_{(1-a^2)^{\frac{1}{2}}} t)}{\phi(D_{(1-a^2)^{\frac{1}{2}}} t)}$$

Let $w = D_a t$ and $r_i = a_i^{-1}(1 - a_i^2)^{\frac{1}{2}}$ then

$$(2.12) \quad h(w) = \frac{\phi'(w)}{\phi(w)} = D_{r^{-1}} \frac{\phi'(D_r w)}{\phi(D_r w)}.$$

By successively setting $w = t$, $r_i = a_i^2$ and $w = D_a t$, $r_i = a_i$ in (2.12)

$$(2.13) \quad h(t) = D_{a^{-2}} \frac{\phi'(D_{a^2} t)}{\phi(D_{a^2} t)} \text{ and } h(D_a t) = D_{a^{-1}} \frac{\phi'(D_{a^2} t)}{\phi(D_{a^2} t)}$$

From (2.13) it now follows that

$$(2.14) \quad h(D_a t) = D_a h(t) \text{ for all } a_i (|a_i| < 1) \text{ and } h(\cdot) \text{ continuous.}$$

A general solution for (2.14) is therefore

$$(2.15) \quad h(t) = At \text{ where } A : p \times p$$

is a diagonal constant matrix (cf. Aczel (1966), p.348).

By using (2.12) and (2.15) it follows that

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$$(2.16) \quad \phi(t) = ce^{\frac{1}{2}t'At}$$

Since $\phi(0) = 1$ and $\left. \frac{\partial^2}{\partial t \partial t'} \phi(t) \right|_{t=0} = -I_p$ it follows that

$$(2.17) \quad \phi(t) = e^{-\frac{1}{2}t't}$$

It follows from (2.8) and (2.17) that

$$\psi(s, t) = \exp\left[-\frac{1}{2}(s's + 2s'D_a t + t't)\right]$$

so that z has a $MN\left(2p, 0 \begin{bmatrix} I_p & D_a \\ D_a & I_p \end{bmatrix}\right)$ distribution.

REMARK 2.3

Much research has been done in the literature on the characterization of the multivariate normal distribution where the vector variates x and $y : p \times 1$ are independently distributed. Khatri and Rao (1977) characterize the multivariate normal distribution based on the characterization of linear functions of independent vector variates. Khatri and Rao (1978) and Khatri (1979) consider further a similar problem of linear functions of more than two dependent vector variates.

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